



there's no way you can interfere with the multiplier, unless you want to decrease it.

The second way (and the method we've used for the purpose of this article) is to adjust the FSB. A good overclocking board will allow increments of the FSB from the default 266 MHz (333 MHz for the newer Intel processors) all the way to 600 MHz.

We tweaked the FSB up from a default of 333 MHz to an overclocked 400 MHz. The processor which was happily running at 2.33 GHz is now *forced* to run at 2.81 GHz, a cool overclock of 477 MHz.

While an overclock of 10 per cent (FSB) isn't a big deal, anything more could result in restarts and freeze-ups. Do not make more than 1 MHz increments at a time to the FSB, just to be safe. In case your CPU refuses to boot into Windows you'll know that you've been too adventurous—tone things down a bit. Once you get Windows to boot, remember to run a CPU intensive benchmark like Super Pi, or WPrime just to make sure your processor is stable at the new clock speed.

Over-volting

After a certain overclock you will find your PC becoming unstable. This is due to the processor running out of power for which you will need to increase the voltage supplied to it. We must caution that over-volting, if done too adventurously, will definitely damage components—so be careful. But if done properly you can further increase clocks. Most motherboards will allow very fine increments to the CPU voltage to the tune of 0.05v.

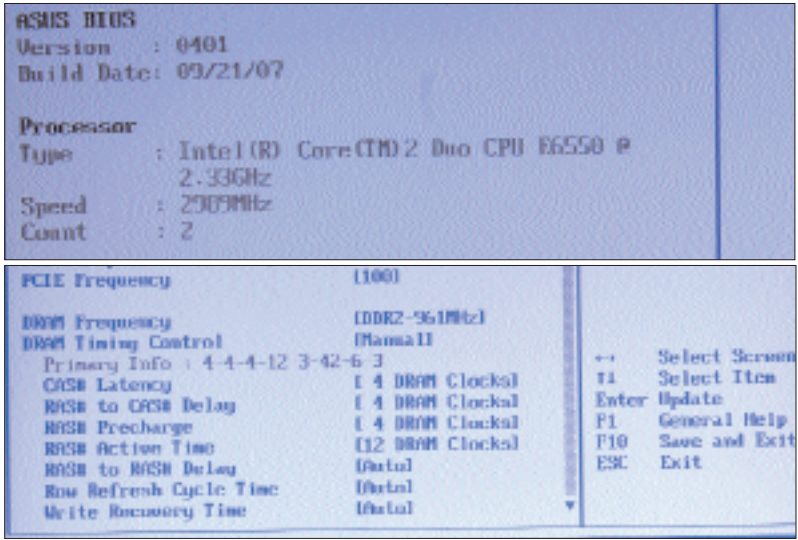
Memory Overclocking

While memory overlocks will not yield as much a performance hike as a CPU overclock, it's all about components working in tandem for overall system performance.

Remember that your CPU is fed by your memory and the faster the memory the quicker

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the CPU is fed. While overclocking memory, it's not just raw MHz increments we're looking at. In fact tighter timings can make all the difference. Typical timings at 800 MHz are 4-4-4-12 at the default 1.8 volts. We however got our memory to go up to 800 MHz at a timing of 3-3-3-8 at 2.2 volts. Memory voltages are not as fine tuned as CPU voltages and you can tweak them with increments of 0.25-0.5 volts without much danger. Do be careful at anything beyond 2.1 volts because some memory chips cannot stand the extra current. However, enthusiast memory (such as the sticks we used) can stand voltages of up to 2.4



Top: The E6550 runs at nearly 3 GHz at an FSB of 400 MHz
Above: Notice how the memory speed jumps to 961 MHz at an FSB of 400 MHz

Tips

- A good practice is to start small and step up gradually. Don't overclock aggressively and be forced to reduce speeds due to heat, instability, freeze-ups etc. Use 1 MHz increments for FSB, after an initial overclock of say 10 per cent.
- Keep an eye on heat. Make sure to run a couple of benchmarks like WPrime, Super Pi and even PC Mark 2005 to see how well overclocked components do under stress.
- Remember that overclocking stresses components and reduces their life. The important thing is to keep them cool especially if you plan on a permanent overclock, or your PC stays on for extended periods. Invest in a good CPU cooler and memory cooler like out Ultra 120 and Dominator memory airflow cooler. Just as a good carpenter needs good tools to show off his craft, an overclocker needs tools to keep his PC cool.
- It may be necessary to add fans inside your cabinet after overclocking as the ambient temperature inside the case may rise. We recommend 2 inlet and 2 outlet fans—all 120 mm.
- Once you reach the maximum stable overclock your component is capable of underclock by 5-10 MHz as a safety margin. Some motherboards have support for overclock profiles in their BIOSes. This is good as one profile can be set at stock speeds and when you want an overclocked rig just select the overclocked profile.

volts. We didn't test this however and wanted to play safe—there's a fine line between speed and recklessness remember!

Since our memory was rated at 1066 MHz you may wonder why we're running it at 800 MHz. The fact is that we ran a few bandwidth tests and found that at 800 MHz (3-3-3-8) our memory gave higher scores than at 1066 MHz (4-4-4-12)—both settings were achieved at 2.2 volts.

Overclocking Graphics

Overclocking a VGA card will only benefit you as far as upping the frame rates in 3D applications, mainly games. When overclocking components we don't generally overclock graphics cards, because the performance impact is minimal in general applications, and the thermal limits of GPUs don't really provide a lot of head room for overclocking.

For ATI, use the ATI Tray Tool, while for NVIDIA we recommend Riva Tuner. When overclocking the core you can use 5 MHz increments initially, but after an overclock of 25 MHz or so, tone down to 1 MHz increments.

When overclocking memory be a little more careful as most graphics cards' memory doesn't really have much headroom. Use 1 MHz increments. Make sure to run a couple of 3D benchmarks like 3D Mark 2006, or loop it a couple of times to verify stability. ■

michael_browne@thinkdigit.com